Predicting Race Outcome Deviations in Formula 1

A PROJECT REPORT

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A Comparative Study on Performance Deviation Classification in Formula 1 Using Distance-Based and Supervised Models

Abstract —

Predicting race outcomes in Formula One (F1) is inherently challenging due to the high variability introduced by strategy decisions, mechanical reliability, and stochastic race events. Traditional models that focus on forecasting winners or final positions often suffer from instability and poor generalization. This study reframes the prediction objective by modeling **performance deviation**, defined as the difference between a driver's qualifying position and final race position.

The problem is formulated as a supervised classification task, where race performances are categorized as outperforming, neutral, or underperforming relative to expectation. Historical F1 race data is transformed into structured feature vectors incorporating qualifying position, historical performance trends, constructor reliability, and deviation statistics. Distance-based similarity analysis using Minkowski metrics is conducted to study sensitivity to feature variations. A comparative evaluation of k-Nearest Neighbors, Logistic Regression, Support Vector Machines, and Random Forest classifiers is performed under multiple parameter configurations.

Experimental analysis investigates the impact of distance metric selection, neighborhood size, and model complexity on classification stability. Bias–variance trade-offs are examined through systematic variation of k in k-NN. Model disagreement analysis is further employed to identify uncertain race scenarios, and K-Means clustering is applied to uncover structural patterns within ambiguous instances.

Results demonstrate that moderate distance sensitivity and nonlinear ensemble methods provide improved stability in deviation classification. The study highlights the importance of metric selection and uncertainty analysis in modeling high-variance competitive systems. Rather than merely predicting outcomes, the proposed framework offers interpretability and robustness insights into performance variability in Formula One racing.

Keywords —

1. Formula 1 Analytics
2. Performance Deviation
3. Supervised Learning
4. Unsupervised Learning
5. Error Analysis

I. Introduction

Formula One (F1) is a highly complex competitive system where race outcomes emerge from interactions between driver skill, constructor reliability, track characteristics, race strategy, and stochastic events such as weather variation or mechanical failures. Traditional predictive models in motorsports focus primarily on forecasting race winners or podium finishes. However, such outcome-based predictions are inherently unstable due to race-day uncertainty and rare high-impact events, making generalization difficult.

This study reframes the predictive objective from winner forecasting to **performance deviation modeling**. Instead of predicting absolute finishing positions, we quantify the deviation between a driver's qualifying grid position and final race position. This formulation transforms the problem into a structured classification task where race outcomes are categorized as:

- **Outperforming expectation**
- **Neutral performance**
- **Underperforming expectation**

By modeling deviation rather than raw outcome, the system reduces noise sensitivity and enables more interpretable performance evaluation.

To investigate this problem, the study integrates:

- Statistical feature engineering of race and constructor attributes
- Vector space representation of driver performance
- Distance-based similarity analysis using Minkowski metrics
- k-Nearest Neighbors classification
- Comparative evaluation of Logistic Regression, Support Vector Machines, and Random Forest
- Bias–variance behavior analysis
- Model disagreement detection as a proxy for predictive uncertainty
- Unsupervised clustering of ambiguous race instances

The objective extends beyond predictive accuracy. The study aims to analyze how model behavior changes with distance metric selection, neighborhood size, and class boundary overlap, thereby providing insights into robustness, stability, and structural uncertainty in competitive racing environments.

By combining supervised and unsupervised learning within a deviation-based framework, this research contributes a more stable and interpretable approach to performance modeling in high-variance sporting systems.

II. Literature Review

Machine learning techniques have been widely applied in sports analytics to model performance and predict outcomes in uncertain environments. In motorsports, most studies focus on forecasting race winners or final standings using historical performance statistics, constructor reliability, and contextual race features. However, such outcome-based predictions are often unstable due to stochastic factors including race incidents, strategy variability, and environmental conditions, which reduce generalization capability.\

Classification-based modeling has been shown to improve interpretability when continuous outcomes are discretized into structured categories (Hastie et al., 2009; Bishop, 2006). This principle supports deviation-based labeling, where performance is evaluated relative to expectation rather than absolute finishing position.

Distance-based learning methods such as k-Nearest Neighbors rely on similarity measures within feature space. Prior research in clustering and metric learning (Jain, 2010; Aggarwal, 2014) demonstrates that distance metric selection significantly influences neighborhood structure and class separability. Minkowski distance provides a generalized framework for analyzing sensitivity to feature variation. However, its systematic evaluation in motorsport performance modeling remains limited.

Ensemble methods like Random Forest (Breiman, 2001) and margin-based classifiers such as Support Vector Machines (Vapnik, 1998) have demonstrated robustness in handling nonlinear relationships and overlapping class distributions. Comparative analysis of these models provides insight into structural complexity within the dataset.

Although prior work addresses outcome prediction and classification theory, limited research explores performance deviation modeling in Formula One using an integrated framework combining supervised learning, distance metric experimentation, bias–variance analysis, and clustering of uncertain instances. This study addresses that gap by systematically evaluating model behavior and uncertainty in deviation-based race performance classification.

III. Methodology

A. System Overview

The objective of this phase of the Formula One performance deviation project is to analyze similarity and classification behavior using statistical and distance-based techniques. Instead of directly predicting race winners, this project focuses on identifying whether a driver **outperforms, underperforms, or performs as expected** relative to their qualifying position.
Lab-3 concepts were applied to historical Formula One race data by converting race records into numerical feature vectors, analyzing statistical properties of driver performance, computing similarity using distance metrics, and applying k-Nearest Neighbors (k-NN) classification to categorize race outcomes.

B. Flow Diagram

The flow diagram represents the overall processing pipeline followed in the proposed system.

Input F1 ace & Qualifying Data
↓
Data Cleaning & Preprocessing
↓
Feature Engineering
↓
Deviation Label Generation
↓
Supervised Classification
↓
Model Disagreement Detection
↓
Unsupervised Clustering
↓
Performance Interpretation

C. Architecture Description

Input Module:
Collects ancient race consequences, qualifying positions, constructor details, circuit data, and race popularity from a public components 1 information source.
Preprocessing Module:
Handles lacking values, filters DNFs wherein required, encodes specific variables, and scales numerical functions to make sure truthful assessment.
characteristic Engineering Module:
Computes race outcome deviation and ancient facts along with common end role, rolling deviation, DNF rate, and constructor reliability.

Deviation Labeling Module:
Converts deviation values into 3 instructions: Outperform, neutral, and Underperform.

Supervised Module:
Trains multiple category fashions including Logistic Regression, help Vector device, and Random woodland to predict deviation classes.

Confrontation analysis Module:
Identifies race instances wherein special fashions expect distinct deviation classes, indicating uncertainty.
Unsupervised Module:
Clusters disagreement samples to find out hidden styles together with risky circuits or method-touchy races.
choice Module:
Produces final analytical insights and model overall performance summaries.

D. Parameter Selection and Justification

Parameter	Value	Justification
Deviation Threshold	+2 positions	Reduces noise from minor position changes
Number of Classes	3	Outperform, Neutral, Underperform
Classification Models	LR, SVM, RF	Covers linear and non-linear behavior
Clustering Algorithm	K-Means	Simple and interpretable
Number of Clusters	3–5	Balances detail&clarity

E. Feature Representation and Vector Operations (F1)

In the F1 dataset, each race entry was treated as a vector containing features such as qualifying grid position, average finishing position, constructor reliability, and deviation from expected performance. Dot product and Euclidean norm were implemented to understand similarity and magnitude of driver performance vectors. These operations helped in understanding how closely two race performances or two drivers resemble each other in multi-dimensional feature space. The correctness of the implementations was verified using NumPy functions.

F. Minkowski Distance Analysis (F1)

Minkowski distance was computed between selected F1 race vectors for values of p ranging from 1 to 10. The variation in distance values demonstrated how sensitivity to feature differences increases with higher values of p.
This experiment highlighted how distance metric selection directly affects neighbor selection in k-NN classification of F1 race outcomes.

G. k-NN Classification (F1)

The F1 dataset was divided into training and testing sets using a two-class deviation setup. A k-NN classifier with $k = 3$ was trained to classify races into outperform or underperform categories.
Prediction accuracy and confusion matrices were analyzed to evaluate classification effectiveness.

IV. Results Analysis & Discussion

A. Intraclass Spread and Interclass Distance in F1 Performance Deviation

Intraclass spread and interclass distance were analyzed to evaluate how well the deviation-based performance classes in the Formula One dataset were separated. The classes considered were **Outperform** and **Underperform**, derived from the difference between qualifying position and final race position.

The intraclass spread represented the variability in driver performance within the same deviation class. Results showed that drivers classified as outperforming exhibited relatively higher spread, indicating that exceptional performances occur under varying race conditions. In contrast, underperforming instances demonstrated comparatively lower spread, suggesting consistency in negative deviation patterns.

B. Feature Distribution and Density Observations

Histogram analysis of selected F1 features, such as qualifying position and deviation magnitude, revealed non-uniform distribution patterns. Mid-grid qualifying positions showed higher variance in deviation, reflecting greater uncertainty in race outcomes due to overtaking opportunities and strategic variations.

Front-grid positions demonstrated lower variance, indicating more stable performance, while back-grid positions showed limited positive deviation potential. These distribution patterns influenced distance calculations and neighbor selection during classification.

C. Behavior of Distance Metrics on F1 Race Data

The Minkowski distance metric was evaluated for values of p ranging from 1 to 10 using selected F1 race vectors. Lower values of p resulted in smoother distance variation, while higher values amplified feature differences.
It was observed that excessively large values of p increased sensitivity to minor feature variations, which negatively affected neighbor stability. Moderate values of p provided a balanced representation of performance similarity, making them more suitable for F1 race classification tasks.

D. k-Nearest Neighbors Classification Results

A k-Nearest Neighbors classifier with $k = 3$ was applied to classify F1 race instances into outperform and underperform categories. The classifier achieved consistent accuracy across both training and testing datasets, confirming its ability to generalize performance patterns rather than memorizing individual race instances. Prediction results showed that most misclassifications occurred near class boundaries, particularly for drivers whose finishing position closely matched their qualifying position. This highlights the inherent uncertainty present in tightly contested races.

E. Effect of k on Classification Performance

The value of k was varied from 1 to 11 to study its impact on classification accuracy. When $k = 1$, the classifier exhibited high sensitivity to individual race outcomes, leading to overfitting. This resulted in higher training accuracy but lower testing accuracy.

As k increased, the decision boundary became smoother, reducing sensitivity to noisy race events. However, very large values of k caused underfitting by oversmoothing meaningful deviation patterns. An intermediate value of k achieved balanced performance, indicating an optimal trade-off between bias and variance.

F. Model Fit Interpretation: Overfitting and Underfitting

Model fit was analyzed by comparing training and testing metrics across different values of k. Overfitting was observed at lower values of k due to excessive reliance on individual race instances. Underfitting occurred at higher values of k, where meaningful deviation trends were lost.
For moderate values of k, the classifier demonstrated stable performance, indicating a regular fit. This balanced behavior is desirable for real-world F1 performance analysis, where robustness to noise and variability is essential.

G. Class Separability and Distance-Based Behavior

Based on intraclass spread and interclass distance observations, the deviation-based classes in the F1 dataset are reasonably but not perfectly separated. Extreme outperform and underperform cases form distinct regions in feature space, while instances close to neutral deviation introduce overlap. This overlap explains misclassifications near class boundaries and reflects the inherent uncertainty present in closely contested races.

The behavior of distance metrics further influences classification. As neighborhood size increases, sensitivity to individual race instances decreases, resulting in smoother decision boundaries. However, excessive smoothing reduces the model's ability to capture meaningful local deviation patterns.

H. Effect of k on k-NN Performance

The performance of the k-NN classifier varies significantly with the choice of k. Smaller values of k lead to high sensitivity to individual race outcomes, capturing noise along with meaningful patterns. This results in higher training accuracy but reduced testing performance. As k increases, the classifier becomes more stable by reducing the influence of outliers.

However, very large values of k dilute local deviation information, leading to reduced classification accuracy. An intermediate value of k provides a balanced representation of neighborhood influence, achieving improved generalization performance.

I. Model Fit, Overfitting, and Suitability of k-NN

Overfitting is observed when k is small, as predictions rely excessively on nearby individual samples. Underfitting occurs at higher values of k where meaningful deviation trends are averaged out. For moderate values of k, training and testing performances are closely aligned, indicating a regular fit.
Based on these observations, k-NN is a suitable classifier for deviation-based F1 performance analysis. When applied with appropriate feature scaling and parameter selection, it effectively captures performance patterns while maintaining generalization capability.

XVI. Conclusion

This study redefined Formula 1 performance modeling by shifting focus from winner prediction to deviation classification. Through systematic experimentation with distance metrics, supervised classifiers, and clustering techniques, the research demonstrated that performance deviation contains both structured and stochastic components. Minkowski distance analysis highlighted the importance of metric selection in neighborhood stability. Bias–variance evaluation confirmed theoretical learning behavior in k-NN classification. Supervised model comparison revealed that nonlinear models better capture deviation complexity.
Model disagreement analysis provided a novel interpretation of race uncertainty, and clustering of ambiguous instances exposed hidden structural influences. Overall, this work illustrates how machine learning systems should be evaluated not only by predictive accuracy but also by robustness, interpretability, and behavior under uncertainty.

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